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Subject : .....

What is CNN?

→ problem → standard neural networks flatten 2D image into 1D array (distroys spatial information)

→ sol → convolutional layers

↳ maintains 2D matrix structure

↳ uses filters (kernals)

What is filter?

→ small matrix of numbers

→ hardcoded filters used to alter images

→ filter applied across entire image pixel by pixel

How filter works in DL?

→ in CNN → not hard coded

→ numbers inside filter matrices → are weights

→ CNN starts with random values of its filters

→ network learns the optimal filter values

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Subject :

What is Convolution?

→ mathematical operation that mixes two lists of numbers or functions to produce third new list

Convolutions in image processing (2D)

→ we slide small 2D matrix of numbers called kernel across all pixels of image

CNN architecture & classification

Structure : → combines feature extraction and classification
→ Before flattening the data into 1D array
built alternates between CL, MPL

advantages : → automatic feature extraction
→ preserve spatial relations in images

Code : from tensorflow.keras import layers, models

```
cnn = models.Sequential([
    layers.Conv2D(...)
    layers.MaxPooling2D(...)
    layers.Flatten(),
    layers.Dense(...)
])
```

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Stride : How many pixels filter shifts H or V across image

Padding : sliding filter over image shrinks output map

↳ "valid" or "same"

Data Augmentation :-

Problem → overfitting

Sol → Data Augmentation expands training dataset by applying random geometric trans

→ Random Flip

→ Random Rotation

→ Random Zoom

Transfer Learning :- → Take pretrained model

→ repurpose weight to specific task

model.trainable = False (Freeze)

image classification

→ assign single label to entire image

Object Detection :

→ identify multiple objects within same image and draw bounding box around one or more

image segmentation :

→ instead of drawing box → create outline and shapes of target